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Advanced Topics in Robotics —Swarm Robotics

Session 2

Literature

An overview of the Boids flocking algorithm [`https://vanhunteradams.com/Pico/Animal_Movement/Boids-algorithm.html`]: This is a brief introduction to the Boids algorithm for flocking. Keep in mind that it was originally developed for computer graphics and animation, so it needs to be adapted for use in robotic systems.

A minimalist flocking algorithm (optional, but recommended) This paper introduces a swarm flocking algorithm designed for robots with minimal capabilities, for example, robots that cannot know the orientation of their neighbors.

An overview on optimal flocking (optional) If you would like to see a bit more on flocking algorithms, this paper is worth reading. The math is not too complicated, but if you don't like it, you can skip it and just focus on the overall ideas the paper is talking about.

Assignment Part 2: Flocking Behavior

The goal of this assignment is to implement a decentralized algorithm that causes the robots to flock. As in the previous part of the assignment, you must implement a controller that runs on each robot independently, using only the sensor types and motion model provided by the simulator. Do not introduce new sensing or motion capabilities beyond what is available.

Remember to keep your code simple.

Hint Depending on your flocking algorithm, you may need to use the method `get_relative_heading(...)`. This method takes the heading of another robot (given in the global frame and received via RAB) and converts it into the local frame of the receiving robot.

Extension 1 Add obstacles and integrate obstacle avoidance into your flocking behavior.

Extension 2 Add one or more light sources, and modulate the behavior based on light intensity (for example, stronger cohesion when light is high, or faster motion in darker areas).

Evaluation and Analysis
You are expected to evaluate the behavior quantitatively. To do so, you must:

- Define performance metrics to measure the effectiveness of the behavior (e.g., alignment, cohesion, number of flocks). You may use available libraries (e.g., DBSCAN from sklearn.cluster for counting flocks).
- Run experiments using these metrics and summarize the results.
- Support your results with relevant statistics based on **multiple repetitions** of each experiment (make sure your runs actually differ, considering the random seed).

In addition to its internal parameters, the performance of your algorithm may also depend on external factors (e.g., swarm size, number and range of sensors). Try adjusting these and analyze how they affect the behavior.

Noise Sensitivity
The simulator includes configurable noise models that are disabled by default (e.g., RAB bearing noise and message dropout, motion noise in position and heading updates, light and orientation sensing noise). Orientation noise is particularly relevant for flocking, since the heading each robot broadcasts comes from its own (noisy) orientation sensor. You are encouraged to experiment with these noise settings to assess the robustness of your algorithm. How sensitive is your behavior to sensing or motion imperfections?

Report A part of your report for this module is about flocking, how it is implemented, how it is evaluated, what parameters and variations you have investigated and how the behavior depends on them. See the workplan for Session 1 for a more detailed description of the report requirements (and feel free to ask if it's not clear to you).